Bag of Visual Words(BoVW) for Image Classification:

Notes:

- The used dataset is **UC Merced Land Use Dataset** with **21** different folders, each folder represents a category with **100** .tif images.

To evaluate the performance of the Bag of Visual Words (BoVW) model for image classification, i varied:

- **Dataset Splits:** The dataset was partitioned into: **50-50, 60-40, 70-30** and **80-20** where each split was stratified to maintain class balance across all 21 land-use categories.

- **Vocabulary Size (K):** The number of visual words in the dictionary was varied across **21, 50, 100, 150, 200, 250** and **300** clusters.

- **Normalization Methods:** Two histogram normalization techniques were tested after visual word quantization: **L1 normalization** and **L2 normalization**.

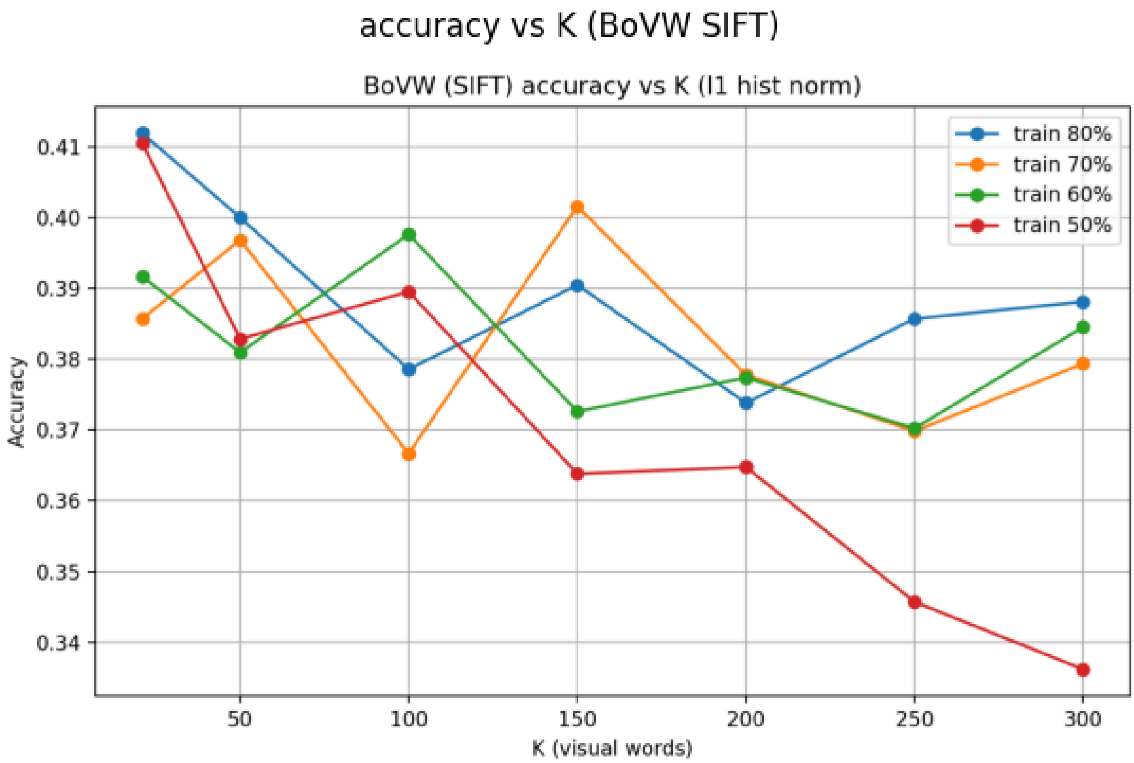
In total, we have 56 different experiments. For each one, SVM classifiers were trained and evaluated. Performance was assessed in terms of **classification accuracy** and **confusion matrices** as we will see.

- In the first experiment we did extract SIFT features from each image on the dataset, compute descriptors and save the result to cache. Then, starting from the second experiment we checked cache and load descriptors directly, we didn't extract SIFT features again.

Obtained results for Normalization = L1:

train\k	**21**	**50**	**100**	**150**	**200**	**250**	**300**
0.5	**0.410476**	**0.382857**	**0.389524**	**0.363810**	**0.364762**	**0.345714**	**0.336190**
0.6	**0.391667**	**0.380952**	**0.397619**	**0.372619**	**0.377381**	**0.370238**	**0.384524**
0.7	**0.385714**	**0.396825**	**0.366667**	**0.401587**	**0.377778**	**0.369841**	**0.379365**
0.8	**0.411905**	**0.400000**	**0.378571**	**0.390476**	**0.373810**	**0.385714**	**0.388095**

The best result was 0.411 with k=21 and training set=0.8.



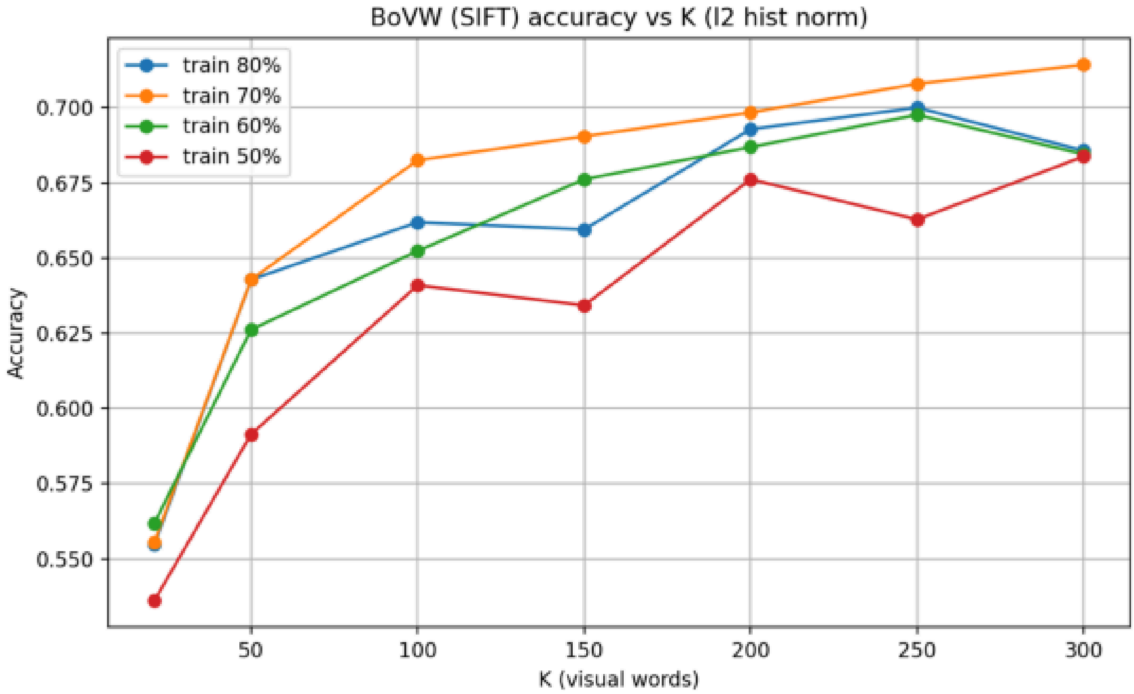
L1 normalization results in weaker and more unstable performance. Across all splits, accuracy remains low and unstable, fluctuating around **0.36-0.41** and increasing the vocabulary size K does not yield consistent improvement. The **L1 normalization** fails to preserve discriminative information in the BoVW histograms because it focuses on relative frequency distribution, it dampens the influence of dominant visual words, which are often critical for class differentiation.

Obtained results for Normalization = L2:

train\k	**21**	**50**	**100**	**150**	**200**	**250**	**300**
0.5	**0.536190**	**0.591429**	**0.640952**	**0.634286**	**0.676190**	**0.662857**	**0.683810**
0.6	**0.561905**	**0.626190**	**0.652381**	**0.676190**	**0.686905**	**0.697619**	**0.684524**
0.7	**0.555556**	**0.642857**	**0.682540**	**0.690476**	**0.698413**	**0.707937**	**0.714286**
0.8	**0.554762**	**0.642857**	**0.661905**	**0.659524**	**0.692857**	**0.700000**	**0.685714**

The best result was 0.714 with k=300 and training set=0.7.

accuracy vs K (BoVW SIFT)



The trend is consistent and smooth, indicating stable learning behavior across configurations, it's because of L2 normalization arises from its ability to preserve the geometric structure of the histogram features:

- It ensures that the histogram vectors lie on the unit hypersphere, making their Euclidean distances directly comparable.
- This aligns perfectly with the assumptions of distance-based classifiers (e.g., SVM, k-NN), improving discrimination between image classes.
- In contrast to L1, which only rescales features by total count, L2 balances the contribution of dominant and rare visual words while maintaining global feature magnitude consistency.

So, L2 normalization enhances the separability of BoVW histograms, leading to higher and more stable accuracy across different K values and data splits which makes the model benefits from both a richer vocabulary (higher K) and a consistent normalization that preserves meaningful feature geometry.

Confusion Matrix confusion_70_300_rep0.png